



Compounds of Transitional elements

81. On heating pyrolusite with KOH in presence of air we get
 (a) $KMnO_4$ (b) K_2MnO_4
 (c) $Mn(OH)_2$ (d) Mn_3O_4
82. $Cu(CN)_4^{2-}$ is colourless as it absorbs light in
 (a) Visible region
 (b) Ultraviolet region
 (c) Infrared region
 (d) All above are wrong
83. Acidified solution of chromic acid on treatment with hydrogen peroxide yields
 (a) $CrO_3 + H_2O + O_2$
 (b) $Cr_2O_3 + H_2O + O_2$
 (c) $CrO_5 + H_2O$
 (d) $H_2Cr_2O_7 + H_2O + O_2$
84. Which of the following metals corrodes readily in moist air
 (a) Gold (b) Silver
 (c) Nickel (d) Iron
85. Which one of the following compounds is not coloured
 (a) Na_2CuCl_4 (b) Na_2CdCl_4
 (c) $K_4Fe(CN)_6$ (d) $K_3Fe(CN)_6$
86. Acidified $KMnO_4$ is decolourized by
 (a) Br_2 (b) O_3
 (c) HCl (d) HBr
87. Which of the following compound expected to be colourless
 (a) ScO (b) V_2O_3
 (c) $CuCN$ (d) $Cr_2(SO_4)_3$
88. Crystals of which pair are isomorphous
 (a) $ZnSO_4, SnSO_4$
 (b) $MgSO_4, CaSO_4$
 (c) $ZnSO_4, MgSO_4$
 (d) $PbSO_4, NiSO_4$
89. On heating $Mn(OH)_2$ with PbO_2 and conc. HNO_3 pink colour is obtained due to the formation of
 (a) $KMnO_4$ (b) $HMnO_4$
 (c) $Pb(MnO_4)_2$ (d) $PbMnO_4$
90. Which of the following is used as white pigment
 (a) TiO_2 (b) V_2O_5
 (c) CuO (d) HgO
91. Which metal oxide is used to make blue glass
 (a) Fe_2O_3 (b) CoO
 (c) Cu_2O (d) NiO
92. The chemical formula for certain pyrophosphate is $Ca_2P_2O_7$, the formula of its ferric pyrophosphate will be
 (a) $Fe_2(P_2O_7)_3$ (b) $Fe_4(P_4O_{14})$
 (c) $Fe_4(P_2O_7)_3$ (d) Fe_3PO_4
93. Which of the following compounds does not dissolve in ammonium hydroxide solution
 (a) AgF (b) $AgBr$
 (c) $AgCl$ (d) AgI



94. Which of the following is non-stoichiometric
(a) Fe_3O_4
(b) Fe_2O_3
(c) FeO
(d) All the above
95. Ferrosilicon is used in steel industry as
(a) A flux
(b) Scavenger of hydrogen
(c) A reducing agent
(d) A cutting tool
(e) Alloying agent
96. In the reaction,

$$2KMnO_4 + 16HCl \rightarrow 5Cl_2 + 2MnCl_2 + 2KCl + 8H_2O$$
the reduction product is
(a) Cl_2 (b) $MnCl_2$
(c) H_2O (d) KCl
97. Which one of the following statements is correct
(a) Manganese salts give violet borax bead test in the reducing flame
(b) From a mixed precipitate of $AgCl$ and AgI ammonia solution dissolves only $AgCl$
(c) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution
(d) On boiling a solution having K^+ , Ca^{2+} and HCO_3^- ions we get a precipitate of $K_2Ca(CO_3)_2$
98. Collin's reagent is
(a) MnO_2/HCl
(b) MnO_4/C_5H_5N
(c) $K_2Cr_2O_7/H_2SO_4$
(d) $Cr_2O_3/2C_5H_5N$
99. Which compound has coloured aqueous solution
(a) $Zn(NO_3)_2$ (b) $LiNO_3$
(c) $Co(NO_3)_2$ (d) $Ba(NO_3)_2$
100. Zinc oxide when treated with $NaOH$ solution gives
(a) $Zn(OH)_2$ (b) $ZnCl_2$
(c) Na_2ZnO_2 (d) $Zn(OH)_3$

